MARITIME POWER RESEARCH · CYBERDEFENSE · STRATEGIC STUDIES

Cyber power as a structural variable of maritime power

A normalized multiplicative model for multidomain assessment of maritime power

$$MP_{norm} = [(\ MI \times NP\ ) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/n_{eff}}$$

$n_{eff} = 3 + \gamma + \alpha$ · $MP_{norm} \in (0, 1]$ · weighted geometric mean (OECD–JRC, 2008; UNDP, 2010)

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Manuscript under review – IEEE Access · diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

OUTLINE

Roadmap of the presentation

1	**Problem**	A sound formula with a structural gap; question and hypothesis
2	**Objectives**	General objective and five specific objectives
3	**Theoretical framework**	Power and its instruments · the maritime power tradition · Booth · the institutional framework · gaps
4	**Method**	Analytical-conceptual approach in four stages
5	**Results**	The two-level formula, its verification, its robustness and the hypothetical Colombian scenario
6	**Discussion**	Contributions, theoretical and policy implications, limitations and agenda
7	**Conclusions**	What the article establishes

THE RESEARCH PROBLEM

A sound formula with a structural gap

THE CLASSICAL FORMULATION

$$MP = MI \times NP$$

Maritime power = maritime interests × naval power

Solís Oyarzún (1997) – Latin American systematization
Uribe Cáceres et al. (2016) – adoption for Colombia
National ocean policy (CCO, 2018) – state policy: Medium Oceanic Power
Naval Development Plan 2042 (ARC, 2021) – currency in naval planning

THE CONCEPTUAL GAP

Cyberspace does not appear as a structural domain of maritime power

Navigation, communications, traffic management, ports and surveillance depend on interconnected digital infrastructures whose vulnerability can affect the exercise of maritime power without physical confrontation.

Nor is strategic will formalized as a variable of the system, nor is maritime security disaggregated.

The article updates the formula – it does not replace it – in continuity with the tradition of the field.

EMPIRICAL EVIDENCE

Cyberspace already degrades maritime power

USD 300 M

Cyberattack on Maersk (2017)

Operations disrupted and losses estimated at USD 300 million.

Wesley et al. (2019); Abbatemarco et al. (2024)

+900 %

Breaches in operational technology

A single compromised system can give access to the other onboard systems.

Akpan et al. (2022)

AIS · ECDIS

Critical maritime systems

Documented vulnerabilities in automatic identification and electronic charts.

Martínez et al. (2024)

2017 → 2021

Resolution MSC.428(98) – IMO

Cyber risk must be addressed in safety management systems.

IMO (2017)

Colombia: 928,660 km² of jurisdictional maritime territory, 4,171 km of coastline and eighteen recognized maritime interests (ARC, 2020, 2021).

QUESTION AND HYPOTHESIS

What the article answers

RESEARCH QUESTION

How should the conceptual expression of maritime power be reformulated to incorporate cyberspace as a strategic domain?

HYPOTHESIS

Cyber power is a structural variable whose omission yields an incomplete representation of state capabilities and vulnerabilities.

OBJECTIVES

General and specific objectives

GENERAL OBJECTIVE

To develop an extended formulation of maritime power that integrates cyber power, strategic will and maritime security, with international theoretical scope and an illustration through a hypothetical scenario for Colombia.

SPECIFIC OBJECTIVES

1. Review, through a documented and reproducible search (2014–2026), the antecedents of the formulation and measurement of maritime power and cyber power.
2. Critically analyze the classical formulation MP = MI × NP to establish its content and analytical limits.
3. Build an extended two-level formulation, normalized, with its functional forms declared as modeling choices and operational boundaries between components.
4. Verify its logical-mathematical properties, its sensitivity and its robustness to uncertainty.
5. Illustrate it with a hypothetical scenario for Colombia informed by a single external ranking.

FRAMEWORK · POWER AND ITS INSTRUMENTS

Where maritime power sits

Power: the ability to affect others to obtain the outcomes one wants; it is stated only with its scope and domain (Nye, 2010).

Instruments of national power (NATO, 2023)

Diplomatic	Information	**Military**	Economic

Maritime interests – economic, social and political: what the state seeks at sea.

Naval power – the expression of military power at sea: its military character gives meaning to its other functions (Booth, 1977).

Maritime power = the capacity to create, develop, exploit and defend maritime interests, in peace and in conflict (CCO, 2018): it combines naval power with maritime interests.

Cyber power – obtains outcomes within and beyond cyberspace (Nye, 2010): **it cuts across both components**; that is the premise of the model.

FRAMEWORK · THE TRADITION

A 130-year tradition, and its next link

1890
Mahan
Six conditions of sea power

1911
Corbett
Control of sea communications

1977
Booth
Trinity: military, diplomatic, policing

1997
Solís Oyarzún
MP = MI × NP

2007
Till
System of interconnected elements

2016
Uribe Cáceres et al.
Adoption for Colombia

2026
This proposal
Extended, normalized formula

THREE LITERATURES THE ARTICLE BRINGS TOGETHER

Maritime power
Mahan · Corbett · Castex · Booth · Till · Solís Oyarzún · Uribe Cáceres

Maritime security
Bueger & Edmunds (2024) – IC · TE · BC

Cyber power
Nye (2010) · Bebber (2022) · Cabuya-Padilla et al. (2024, 2025)

FRAMEWORK · BOOTH (1977)

What navies do, and what cyberspace changes

THE TRINITY OF NAVAL FUNCTIONS

Unified by the use of the sea:

- **Military** – the base of the trinity
- **Diplomatic** – foreign policy short of the use of force
- **Policing** – sovereignty over one's own maritime frontiers

They do not weigh equally: each state weighs them by its interests → the model weights are calibrated by state.

LOSS-OF-POWER GRADIENT

Naval power: decays with distance until it ceases to be effective (Booth, 1977, p. 120).



Cyberspace: instantaneous global reach (NATO, 2025, para. 2.25).



The reach of cyber power does not decay with distance, although its effect still depends on access and infrastructure (Nye, 2010): hence it enters as a factor of the whole system, not as part of naval force.

FRAMEWORK · COLOMBIAN INSTITUTIONAL FRAMEWORK

From ocean policy to the naval plan

PNOEC 2016-2030 (CCO, 2018)

State policy: make Colombia a **Medium Oceanic Power**. Frame of the actions of all sectors and input to the National Development Plan and the sectoral plans.

Position **Political will** **Maritime interests**

↓

NAVAL DEVELOPMENT PLAN 2042 (ARC, 2021)

"To protect and promote maritime and fluvial interests through naval power"

18 maritime interests (CCO, 2021). The 13 of the PNOEC included naval power; the list of 18 excludes it and avoids double counting in MI × NP.

Institutional description: maritime power as "the sum" of interests and naval power (CCO, 2021). The model adopts the product: a sum would let the absence of a component be offset.

No mention of cyberspace or cybersecurity in the PNOEC or in the two documents that define the maritime interests (ARC, 2020; CCO, 2021).

FRAMEWORK · STATE OF THE ART

Four gaps within the documented search

1 Cyber power is not formalized

None of the 812 screened records formalizes it as a structural variable of a maritime power measure.

2 Strategic will stays outside the system

It is treated as a driver of naval power, not as a condition of the whole system.

3 Maritime security not disaggregated

Without the threat dimensions of the contemporary agenda or cyberspace.

4 Existing measures are additive

A deficit in one component is fully offset by a surplus in another.

Documented search (OpenAlex + Scopus, 13-oct-2026, 2014–2026): 1,045 records → 812 unique → 13 measure maritime or naval power → **0 include cyber power**. IMEMO, Chairil et al., Crisher and Souva, Zhang et al., Vargas-Suárez et al.: all additive.

METHOD

Analytical-conceptual approach, in four stages

Conceptual formalization precedes quantitative operationalization – the same tradition as Mahan, Corbett, Till and Solís Oyarzún.

1

STAGE 1

Documented search

Four queries in OpenAlex and Scopus; records and screening published.

2

STAGE 2

Critical analysis

Content and limits of the classical formula in its sources.

3

STAGE 3

Building the formula

Two levels; aggregation rules chosen by substitutability; normalization by n_{eff}.

4

STAGE 4

Hypothetical scenario: Colombia

MS rank from Vargas-Suárez et al. (2021); other values assumed.

Verification ≠ calibration ≠ external validation (Sargent, 2011). The model is verified; calibration and external validation are left as an agenda, with a stated protocol.

CENTRAL RESULT · LEVEL 1

The extended formula of maritime power

$$MP_{norm} = [(\ MI \times NP\ ) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/n_{eff}}$$

$$n_{eff} = 3 + \gamma + \alpha \rightarrow MP_{norm} \in (0, 1] \cdot \omega_{MI} = \omega_{NP} = \omega_{SW} = 1/n_{eff},\ \omega_{MS} = \gamma/n_{eff},\ \omega_{CP} = \alpha/n_{eff}$$

MI
Maritime interests
Purpose: what the state seeks at sea

NP
Naval power
Instrument: force, position and MDA

SW
Strategic will
Multiplier of the whole system

MS
Maritime security
Condition of the environment; weight γ

CP
Cyber power
Structural variable; weight α

Keeps the classical core. **Limited substitutability**: a deficit is only partly offset, and the collapse of any component cannot be offset (OECD–JRC, 2008; Greco et al., 2019).

RESULTS · LEVEL 2

Disaggregation of the five components

MI

$$MI = \Sigma\, w_i \cdot MI_i$$

Linear weighted sub-index: interests are substitutable and weights act as substitution rates. Colombia: n = 18 (CCO, 2021).

NP

$$NP = (F \times P \times MDA^{\mu})^{1/(2+\mu)}$$

Weighted geometric mean of force (with tactical-operational cyberdefense), position and maritime domain awareness: the collapse of any factor drives NP to zero.

SW

$$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$$

Political, institutional and social will; geometric mean: none offsets the absence of another.

MS

$$MS = (IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$$

Interstate conflicts, terrorism and blue crime: measures the incidence of threats, not capabilities.

CP

$$CP = \min\{1,\ (\beta_1 CS_m + \beta_2 CD_m + \beta_3 RC_m)(1+\delta \cdot CDip)\}$$

Cybersecurity, strategic cyberdefense and recovery capacity, amplified by cyber diplomacy.

Explicit operational boundaries: each indicator is assigned to a single component, to avoid double counting.

RESULTS · THE CENTRAL CONTRIBUTION

Cyber power, inside

CS_m **Maritime cybersecurity** – Controls, threat analysis and information sharing on the IT/OT assets of the domain.

CD_m **Strategic maritime cyberdefense** – National, non-executional capacity: governance, cooperation, doctrine, science, education and deterrent posture.

RC_m **Maritime recovery capacity** – Speed of restoration: $RC_m = \min\{1, t_{ref}/t\}$.

CDip **Cyber diplomacy** – Amplifier only: never reduces CP; saturation keeps CP ≤ 1.

α

WEIGHT OF CYBER DEPENDENCE

Proposition 1: $MP_{norm} = G^{1-w} \cdot CP^{w}$, $w = \alpha/n_{eff}$.

Raising CP always raises the index; raising α raises it only if CP > G. Concavity holds for **any** $\alpha > 0$.

Colombian scenario: G = 0.464 > CP = 0.40 → a larger α lowers the index (0.464 → 0.450).

Cyber maturity is read on CP itself, in five bands: incipient, basic, consolidating, advanced, consolidated.

RESULTS · INTERPRETATION

A five-level analytical scale

High	(0.80, 1]	Consolidated multidomain projection
Significant	(0.60, 0.80]	Regional power with developed capabilities
Moderate	(0.40, 0.60]	Developing power with identifiable gaps ★ **Colombian scenario: MP_{norm} = 0.455**
Low	(0.20, 0.40]	Incipient capability with structural vulnerabilities
Critical	(0, 0.20]	Structural maritime vulnerability

FOUR USES, ONCE INPUTS ARE MEASURED

- Diagnosis of the current level
- Target level for planning
- Comparison among middle powers (common weights and anchors)
- Scenarios in the simulator

Five equal, contiguous, half-open intervals of the normalized range: conventions of the article, not thresholds estimated from data.

RESULTS · VERIFICATION

Three verification stages

A — Logical-mathematical properties

Collapse, monotonicity, concavity for any $\alpha, \gamma > 0$, bounds. 300,000 draws; maximum deviation 1.2×10^{-10}.

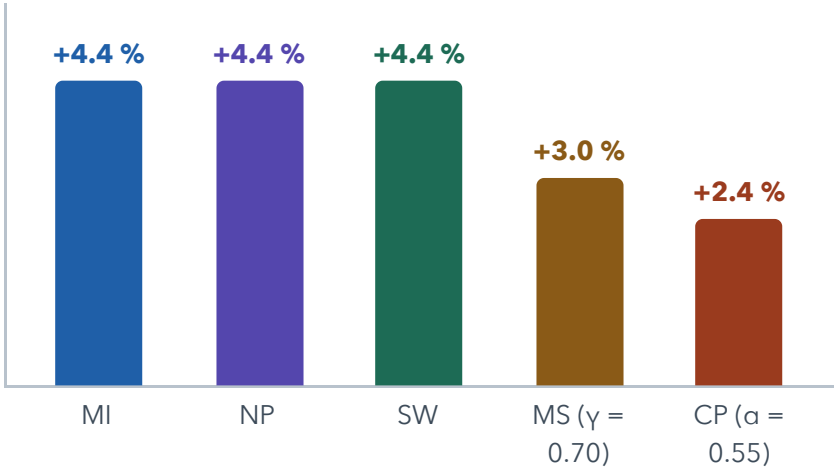
B — Sensitivity (+20%)

The elasticity of each component equals its normalized weight ω_k.

C — Constructed profiles

Critical ≈ 0.19 · reference ≈ 0.52 · consolidated ≈ 0.85: discriminates along the scale.

ΔMP_{norm} for a 20% improvement



Cyber power has the lowest proportional response ($\omega_{CP} = 0.129$). Sensitivity is not efficiency: comparing investments requires costs.

RESULTS · UNCERTAINTY AND ROBUSTNESS

How robust is the result of the Colombian scenario?

	Components ± 0.10	Components ± 0.20
Median MP_{norm}	0.464	0.452
P5–P95 band	[0.406, 0.524]	[0.346, 0.557]
Moderate / Low	96.6 % / 3.4 %	78.1 % / 21.0 %
S_{Ti} of MS · γ	0.204 · 0.361	0.325 · 0.163
S_{Ti} of α	0.035	0.025
MS is the lowest	98.0 %	76.8 %

Monte Carlo with Sobol indices (Saisana et al., 2005; Saltelli et al., 2010); α and γ in [0.10, 1]. Ranges are the authors' assumptions, not estimates.

1 · Moderate is the most likely level

Robust to moderate uncertainty; with wide ranges, Low appears in 21% of the draws.

2 · MS and γ dominate the variance

In this scenario, measuring maritime security and calibrating γ come first, and α is the least influential factor; with freely varying sub-indicators, strategic will and naval power dominate (Table 8).

3 · The aggregation rule does not change the level

Arithmetic mean 0.454 · unweighted geometric 0.433 · MP_{norm} 0.455: all three Moderate.

RESULTS · ILLUSTRATIVE SCENARIO

Colombia: what the model decomposes

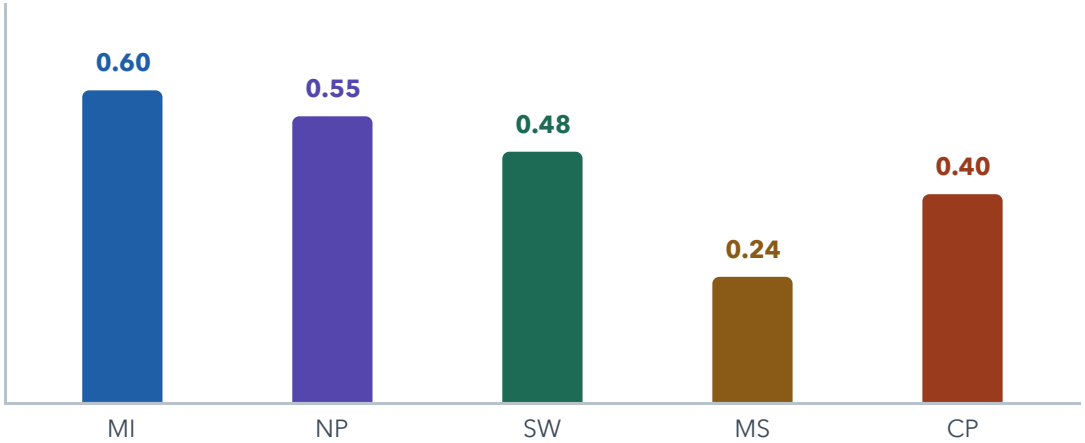
MP_{norm}

0.455

MODERATE LEVEL

Only MS comes from an external source: rank 75 of 99 in maritime security (Vargas-Suárez et al., 2021), converted into a position percentile, (99 − 75)/98 ≈ 0.24. The other values and the parameters are assumptions of the authors, set directly at Level 1.

Scenario values by component (α = 0.55; γ = 0.70)



Maritime security is the lowest component and the one with the largest marginal return; cyber power is the second lowest, at the top of the *basic* band. The values are assumed: they show where measurement should begin, not established gaps.

An illustration, not a validation or a diagnosis: the value is not compared with the Vargas-Suárez et al. index, which is on another scale.

RESULTS · STRATEGIC IMPLICATION

Is cyber power the most efficient investment vector? It depends on costs

Absolute marginal return $\partial MP_{norm}/\partial X_k$

Component	Return
MS – Maritime security	0.312
SW – Strategic will	0.223
NP – Naval power	0.195
MI – Maritime interests	0.178
CP – Cyber power	0.147

With the scenario values, cyber power ranks last in marginal return.

Corollary 1 – cost condition

$$c_k/c_{CP} > (\omega_k/X_k) \,/\, (\omega_{CP}/CP)$$

Cyber power would be the most efficient vector only if raising each other component by one unit cost at least:

1.21×	1.32×	1.52×	2.12×
MI	NP	SW	MS

what it costs to raise cyber power. The model contains no costs: this is a testable condition, not a result.

Sensitivity measures responsiveness, not efficiency: prioritizing an investment requires knowing the relative costs.

DISCUSSION · CONTRIBUTIONS

Six contributions

1	**Cyber power as a structural variable**	With explicit operational boundaries and α as a weight, not a capability.
2	**Strategic will as a multiplier of the system**	It conditions all of maritime power, not only naval power.
3	**Disaggregated maritime security**	IC · TE · BC as incidence of threats, with weight γ.
4	**Normalization, scale and verification**	$MP_{norm} \in (0, 1]$, five levels, Proposition 1 and Corollary 1; verification distinguished from calibration and validation.
5	**MDA ≠ maritime consciousness**	Technical-operational capability in NP; cultural construct in SW.
6	**Maritime power within a general theory of power**	Naval power as the expression of military power; maritime interests as the economic, social and political component; cyber power cross-cutting.

DISCUSSION · IMPLICATIONS

Two implications beyond the formula

THEORETICAL

The reach of cyber power escapes the geographic logic of navies

Classical theory distinguishes navies by a reach that decays with distance (Booth, 1977); cyberspace offers instantaneous global reach (NATO, 2025).

Consequence: for states that relied on their distance from major naval powers, distance no longer guarantees protection against being reached; effectiveness still depends on access, infrastructure and preparation.

POLICY

Ocean policy already counts will, but not cyber

The PNOEC frames the development plans and treats political will as a variable of its goal (CCO, 2018); neither it nor the two documents that define the maritime interests mention cyberspace or cybersecurity, and one of them describes maritime power as a sum.

Consequence: the model offers the structure to incorporate it in the next update and, once calibrated and validated, to track progress with an index in which no collapse is offset.

DISCUSSION · LIMITS AND AGENDA

Stated limitations, an agenda with a protocol

LIMITATIONS

- Verified, but neither calibrated nor externally validated.
- In the hypothetical scenario, only MS comes from an external source; the novelty claim is bounded by the documented search.
- Static model; the enabling effect of cyber power on NP and MS is argued but not represented.
- Robust to moderate uncertainty, not to large uncertainty.

CALIBRATION AND VALIDATION PROTOCOL (NATIONAL INDEX)

1. **Expert calibration:** AHP panel on the five components; γ and α as weight ratios.
2. **Independent inputs:** published sources (ITU, 2024; NCSI; Wild and Fraser, 2026).
3. **Uncertainty:** Monte Carlo with each expert's weights and each source's error.
4. **Comparison:** rank correlation with several benchmarks (Vargas-Suárez et al., 2021; Crisher and Souva, 2026; IMEMO), with the criterion fixed beforehand.

Also: regional extension and dynamic modeling with SERDUX-MARCIM.

CONCLUSIONS

Maritime power is also exercised in cyberspace

01 The classical formula remains valid but is no longer sufficient: the proposal updates it while keeping its multiplicative core.

02 Cyber power, strategic will and maritime security enter as structural variables, with limited substitutability: the collapse of any component cannot be offset.

03 Maritime power combines naval power – the expression of military power at sea – with maritime interests; cyber power cuts across both, and its reach does not decay with distance.

04 The hypothetical Colombian scenario falls in the Moderate level ($MP_{norm} \approx 0.455$), with maritime security as its lowest component; it is an illustration, not a diagnosis.

05 Colombian ocean policy already treats will as a variable but does not mention cyberspace or cybersecurity: the model offers the structure to incorporate them.

06 Whether cyber power is the best investment depends on relative costs (Corollary 1); the national index is the next stage.

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Open-access interactive simulator: diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

Thank you